

Youngjae Cho

ML Research Engineer

Seoul, South Korea | leon5760@gmail.com

youngjae-cho.github.io | GitHub | Google Scholar

I study **robust and data-efficient machine learning**, with a focus on preference optimization, active learning, Bayesian adaptation, and video understanding.

EDUCATION

KAIST | M.S., Industrial & Systems Engineering

Mar 2022 - Feb 2024

Advisor: Il-Chul Moon

KAIST | B.S., Industrial & Systems Engineering

Mar 2017 - Feb 2022

SELECTED PUBLICATIONS

NeurIPS 2026

Accepted

First author

Learning Where It Matters: Geometric Anchoring for Robust Preference Alignment

Youngjae Cho, Jongsuk Kim, Ji-Hoon Kim

Geometric anchoring for robust preference optimization; **+3.6 percentage points** in AlpacaEval 2.0 LC win rate over SimPO.

arXiv 2026

Preprint

First author

Beyond Scalar IoU: Structured Verification from Rollout Groups for Video Temporal Grounding

Youngjae Cho, Won Young Jhoo, Jongsuk Kim

SUTURE: Rollout-group verification for video temporal grounding, improving performance across five benchmarks.

AAAI 2024

First author

Make Prompts Adaptable: Bayesian Modeling for Vision-Language Prompt Learning with Data-Dependent Prior

Youngjae Cho, HeeSun Bae, Seungjae Shin, YeoDong Youn, Weonyoung Joo, Il-Chul Moon

Bayesian prompt adaptation under scarce data and distribution shift.

ICML 2023

Co-first author

Sharpness-Aware Active Learning

Yoon-Yeong Kim*, **Youngjae Cho***, JoonHo Jang, Byeonghu Na, Yeongmin Kim, Kyungwoo Song, Wanmo Kang, Il-Chul Moon

Sharpness-aware acquisition for label-efficient learning.

* Equal contribution. Additional publications appear on page 2.

RESEARCH EXPERIENCE

Pyler

ML Research Engineer

Oct 2025 - Present

Preference optimization. Developed GAPO (NeurIPS 2026, accepted) and multimodal preference-optimization methods for Nemotron-Nano-12B-v2-VL using SimPO and Megatron-Bridge.

Video understanding. Developed SUTURE, a rollout-group verifier for video temporal grounding (arXiv 2026; first author).

Auditable supervision. Recovered decision trees from 299K LLM/VLM reasoning traces; improved content-safety macro F1 from **0.777 to 0.857**.

Aiv Co.

ML Research Engineer

Mar 2024 - Oct 2025

Generative modeling. Led background-aware diffusion research for industrial anomaly detection. Evaluated on MVTec-AD and LOCO and deployed defect synthesis to improve production detector precision and recall.

KAIST

Graduate Research

Mar 2022 - Feb 2024

Data-efficient learning. Developed Bayesian vision-language prompt adaptation (APP, AAAI 2024; first author) and sharpness-aware active learning (SAAL, ICML 2023; co-first author).

ADDITIONAL PUBLICATIONS

arXiv 2024 Preprint First author	Background-Aware Defect Generation for Robust Industrial Anomaly Detection Youngjae Cho, Gwangyeol Kim, Sirojbek Safarov, Seongdeok Bang, Jaewoo Park
ICML 2022 Workshop	Improving Group-based Robustness and Calibration via Ordered Risk and Confidence Regularization Seungjae Shin, Byeonghu Na, HeeSun Bae, JoonHo Jang, Hyemi Kim, Kyungwoo Song, Youngjae Cho, Il-Chul Moon Workshop on Spurious Correlations, Invariance and Stability
IEEE SMC 2021	Predict Sequential Credit Card Delinquency with VaDE-Seq2Seq Yeongmin Kim, Youngjae Cho, Hanbit Lee, Il-Chul Moon

ACADEMIC SERVICE

ICML 2026 | Gold Reviewer

NeurIPS 2026 | Reviewer

AWARDS

NVIDIA Nemotron Hackathon

Track B Winner | 2026

Team winner, Domain-Specialized Model track.